Google Cloud

Introduction to AI and Machine Learning on Google Cloud

Module 2

Generative AI



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01 AI foundations ☑

02 Generative AI ☐

03 AI development options ☐

04 AI development workflow ☐

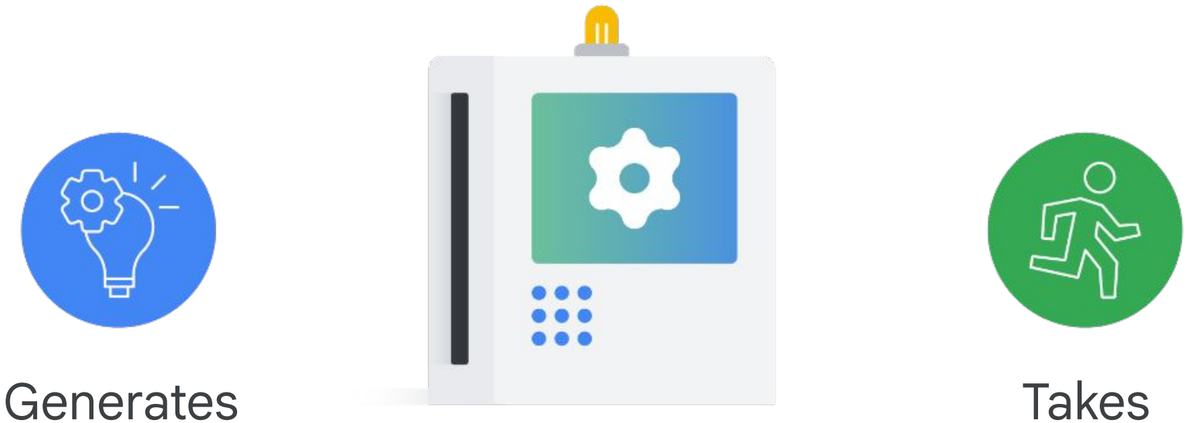
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Generative AI on Google Cloud

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What is generative AI?

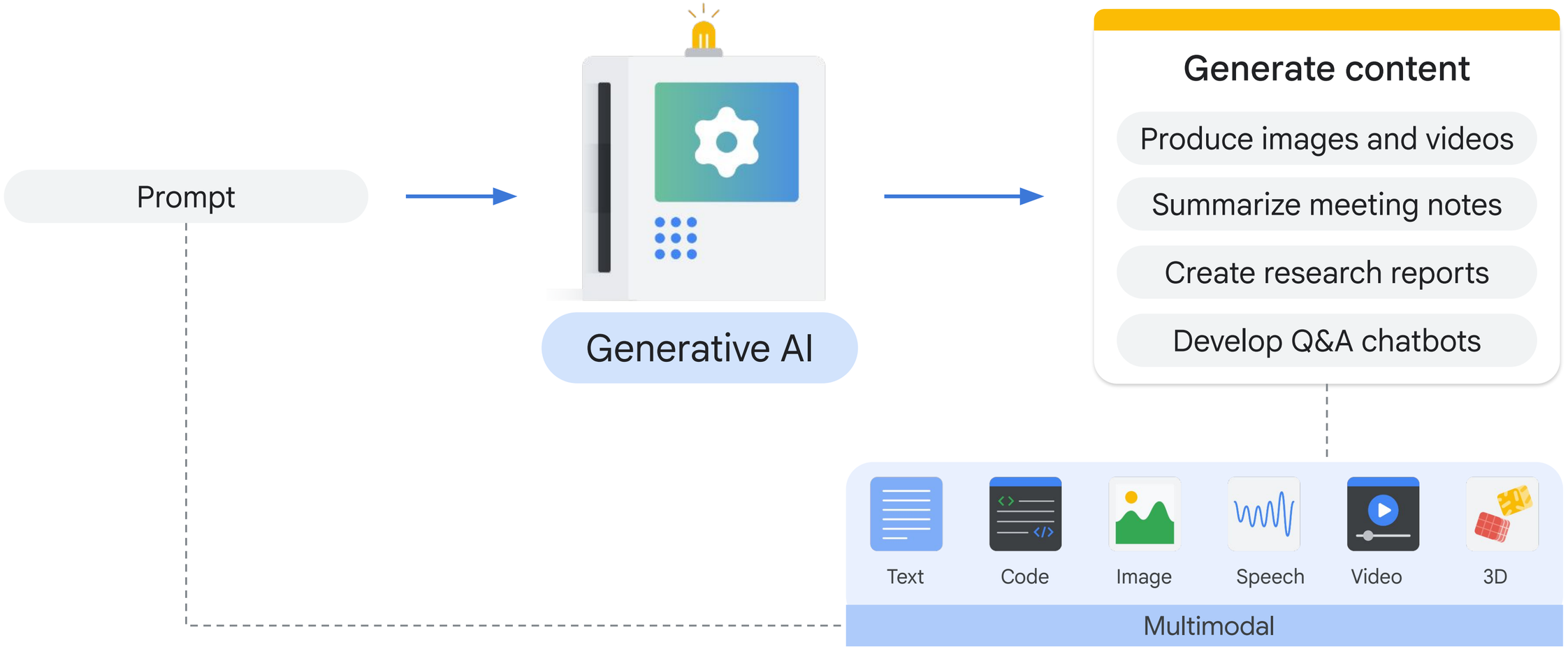


Generates content

Takes action

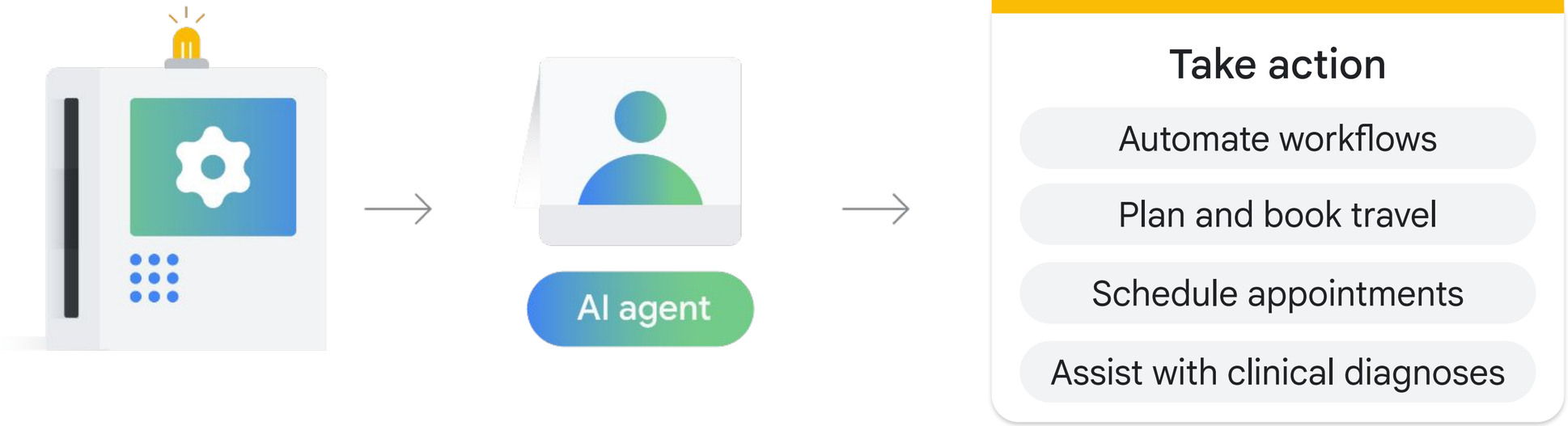
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Gen AI: Generate content



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Gen AI: Take action



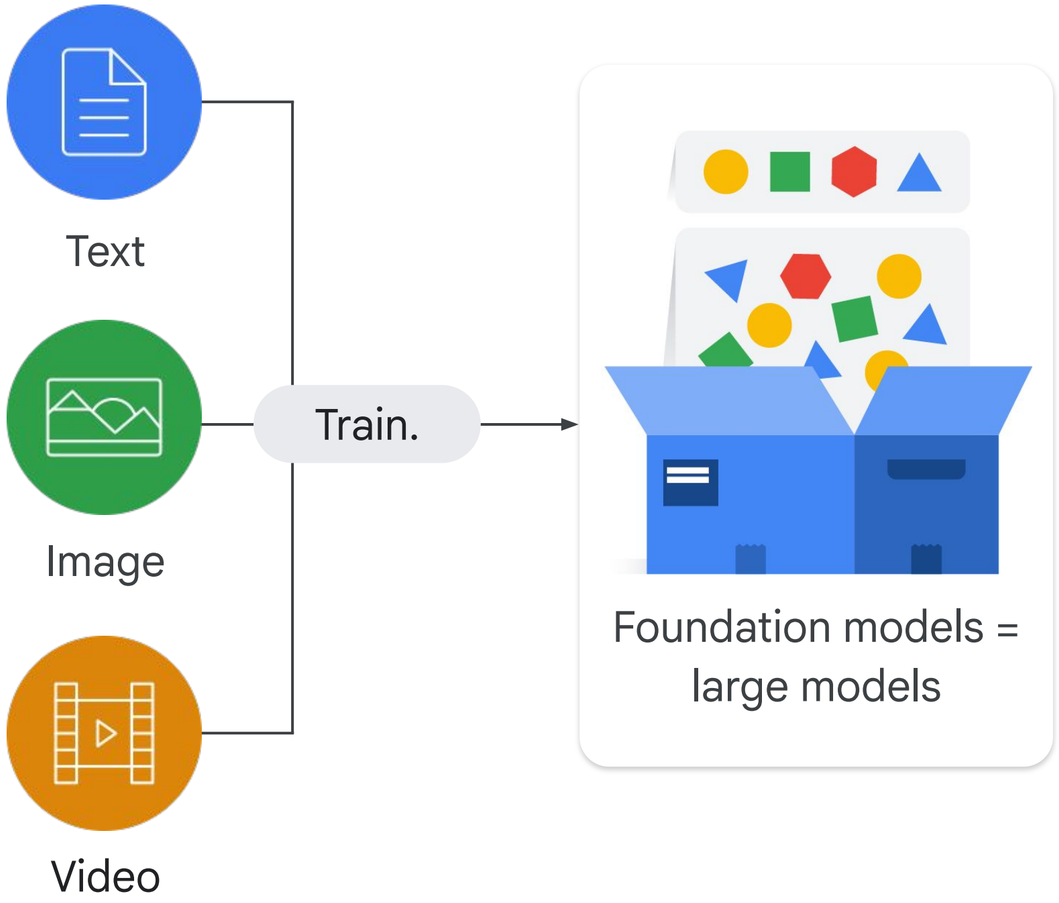
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Foundation models

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How does AI generate content?

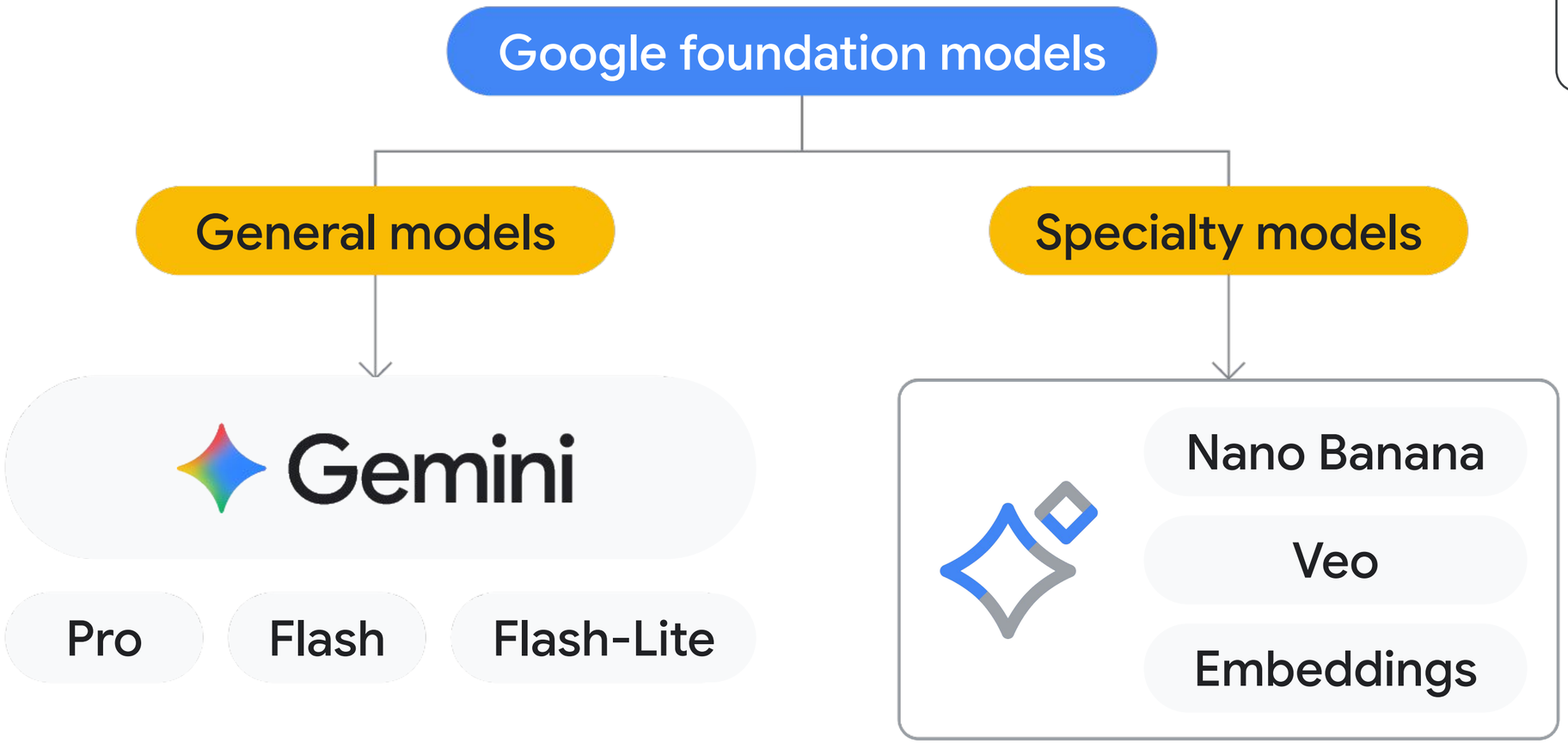


- Significant number of parameters
- Vast training data
- High requirements of computational power

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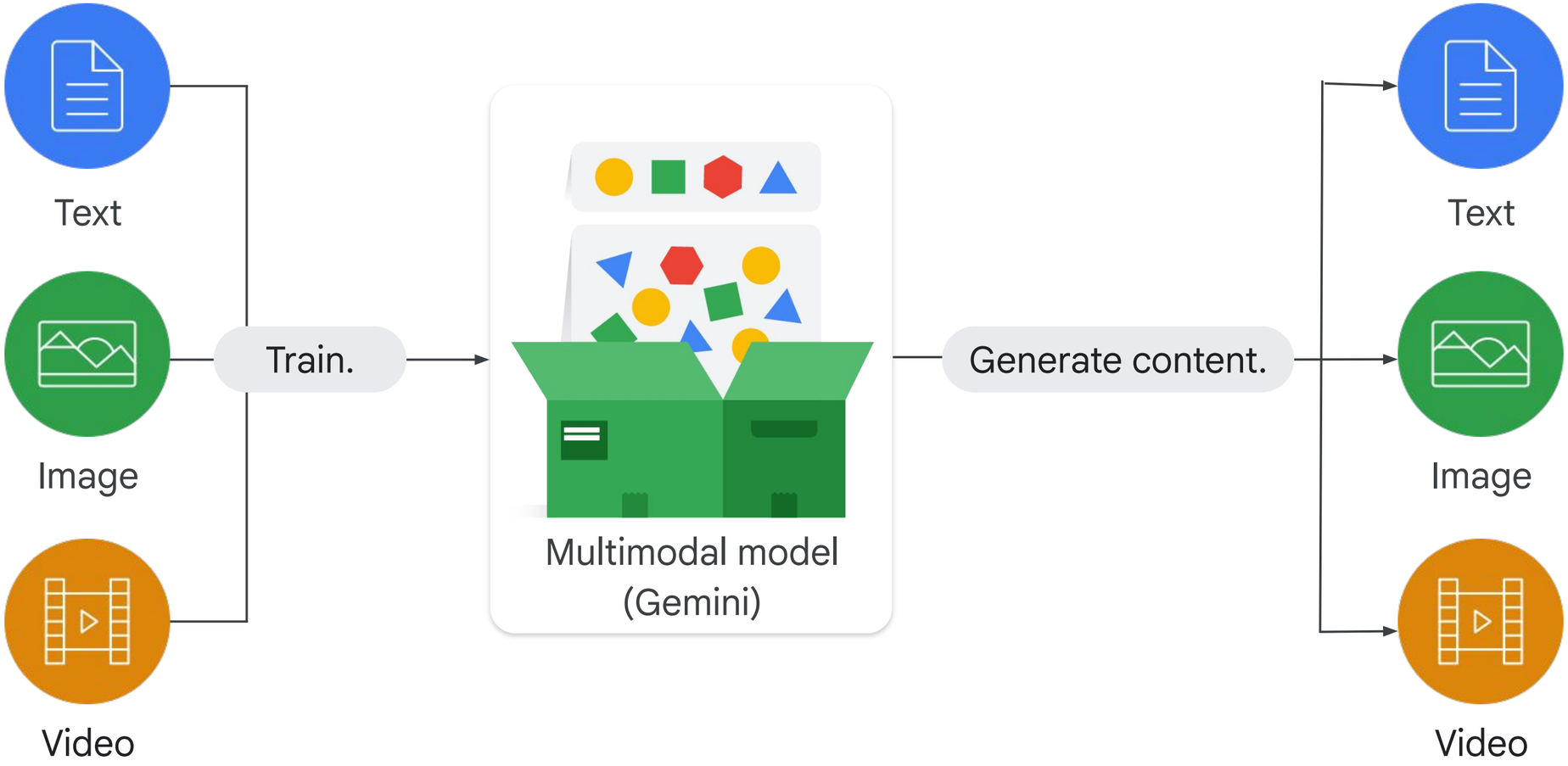
Google’s foundation models

Latest updates



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What is a multimodal model?



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Which foundation model best processes multimodal data?

A Veo

B Embeddings

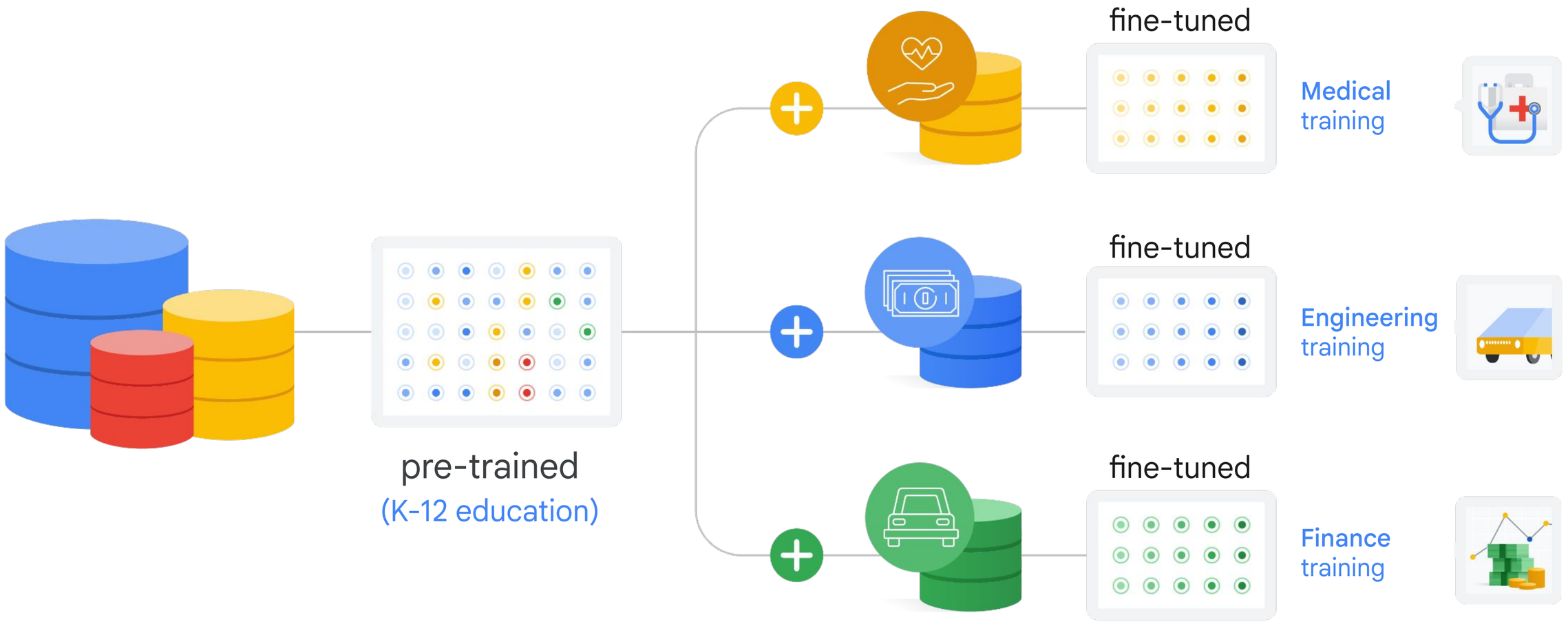
C Nano Banana

D Gemini

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Pre-trained versus fine-tuned models

Use a K-12 (ม.6) analogy



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Idea to app

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Use case and personas



Bea

Business analyst

- Wants to quickly prototype gen AI app.
- Lacks technical background.



Ian

ML engineer

- Requires robust, secure, scalable tool.



Ann

AI developer

- Needs user-friendly development platform.

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Google offers a variety of gen AI products and services



Bea

Business analyst



Ian

ML engineer



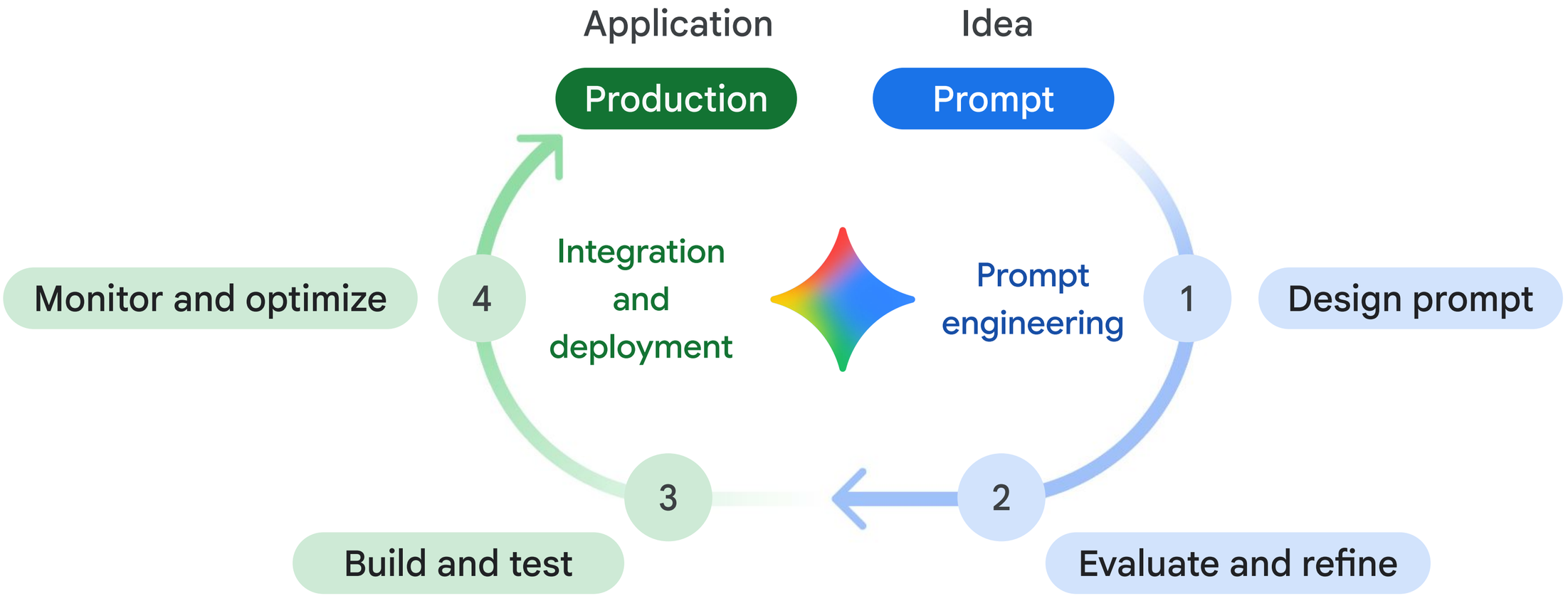
Ann

AI developer

- **Gemini Enterprise app:** Easily discover, create, customize, and orchestrate agent.
- **Agent Platform:** Build, scale, govern, and optimize enterprise-grade AI agents.
- **Gemini Enterprise for CX:** Deploy, manage, and evolve intelligent agents to deliver seamless customer journeys.

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Prompt to production lifecycle



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How to make a “good” prompt?

You need to be good at asking questions.

Prompt anatomy

A prompt includes:

- Task (required)
- Context (optional)
- Examples (optional)

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Task

Task (required)

< >

Conduct a risk analysis for an insurance company.

Tell the AI what you want.



Zero-shot prompting = providing only the task without any examples

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Context

Context (optional)

< >

You are a business analyst overseeing risk assessment for an insurance company.

Set the stage for the AI.



System instructions

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Examples

Examples (optional)

- Desired responses
- Step-by-step instructions
- Output format

Show the AI how to respond.



- Few-shot prompting = Providing a few examples

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Example of a well-structured prompt

Prompt example

Context

You are an IT help desk technician at a university. Your daily job is to help faculty and students solve their technology issues.

Task

To complete the task, you will need to follow these steps:

1.
2.
3.

Examples

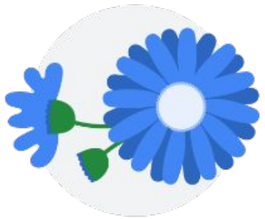
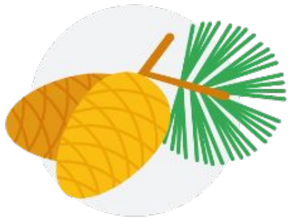
Common Q&A:

Reset password →

Create a new account →

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Temperature



`[flowers (0.5), trees (0.23), herbs (0.05), …, bugs (0.03)]`

Low

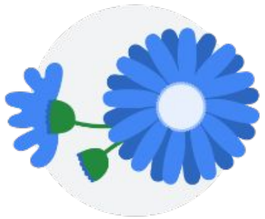
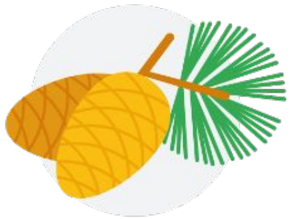
High

- Narrows the range to **high-probability** words.
- Use when you expect a more “typical” answer.

- Expands the range to low-possibility words.
- Use to generate more "creative” content.

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Top K



`[flowers (0.5), trees (0.23), herbs (0.05), …, bugs (0.03)]`

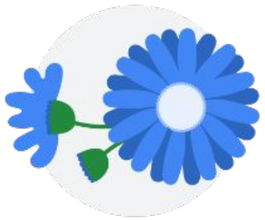
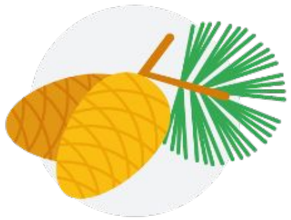
Top K

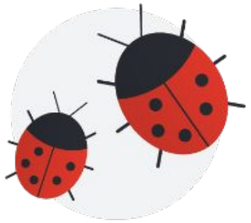
The model returns a random word from a set of top K possible words.

For example: K = 2 flower (0.5) and tree (0.23)

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Top P



`[flowers (0.5), trees (0.23), herbs (0.05), …, bugs (0.03)]`

Top P

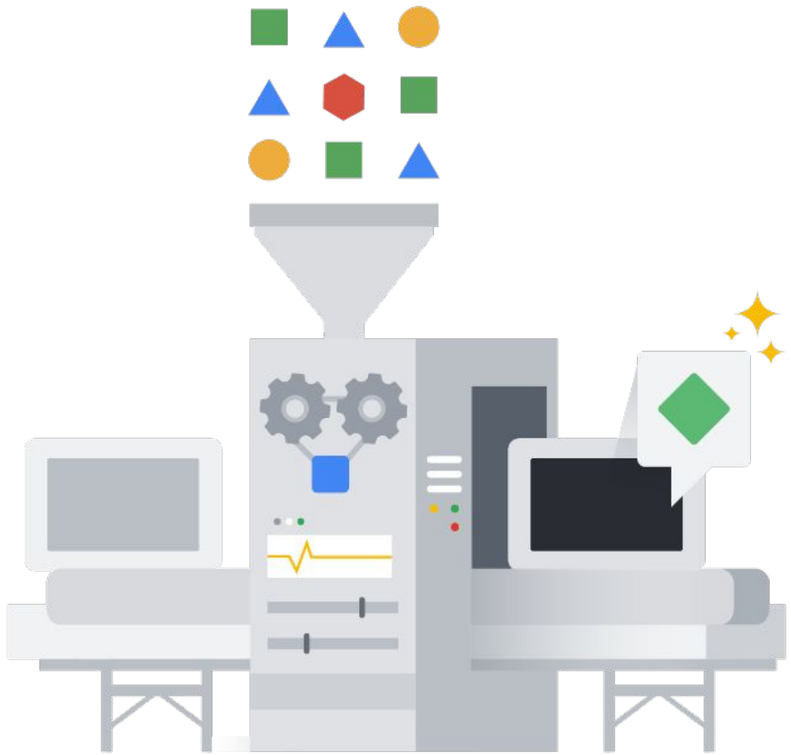
The model returns a random word from the smallest subset with the sum of the likelihoods that exceeds or equals to P.

For example: P = 0.75 Flowers (0.5)+trees (0.23)+herbs (0.05)=0.78 (> 0.75)

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Model parameters

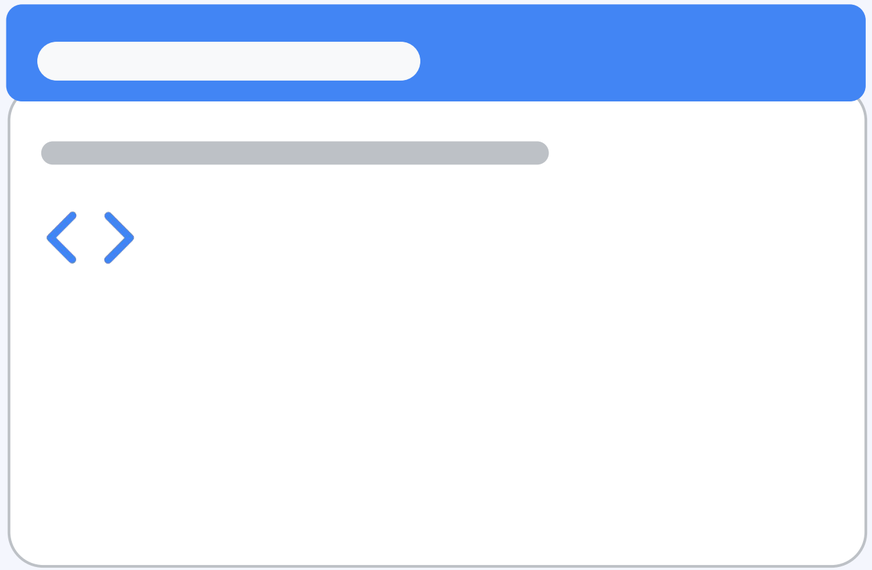
- Model type
- Temperature
- Top K (number)
- Top P (probability)



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Generate evaluation metrics by adding ground truth

Ground truth



Ground truth = preferred answer

- Enables you to generate your own evaluation metrics.
- Responses are evaluated against ground truth answer.

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Deploy an application

User interface (UI)

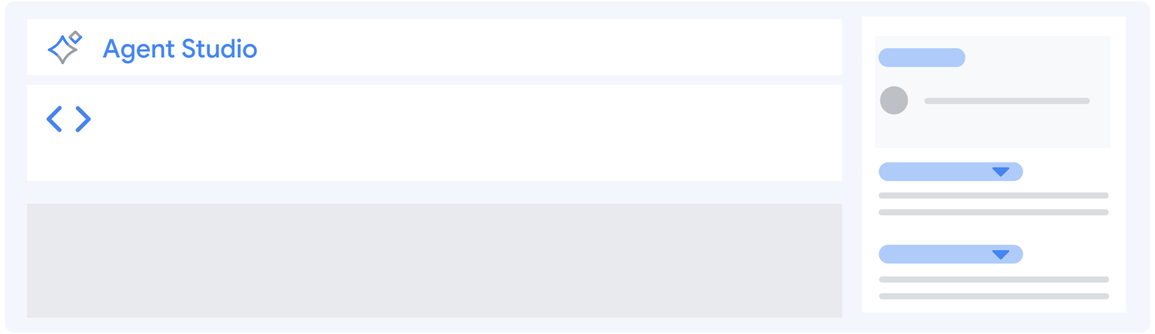
Using the Google Cloud console UI (no-code).

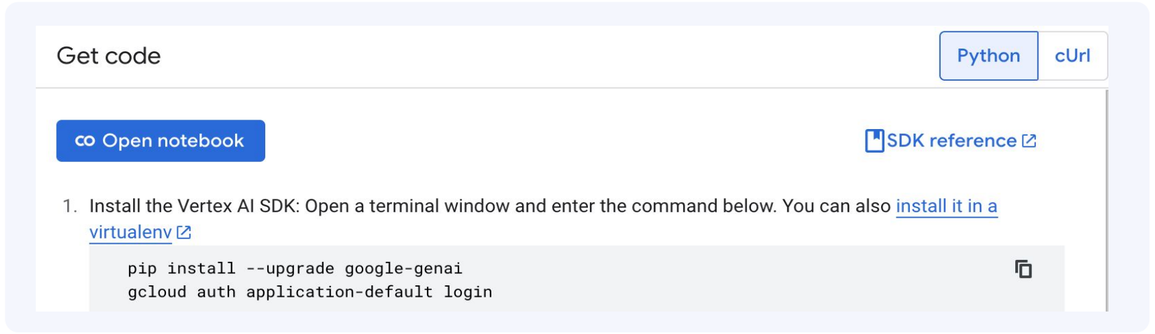
SDKs

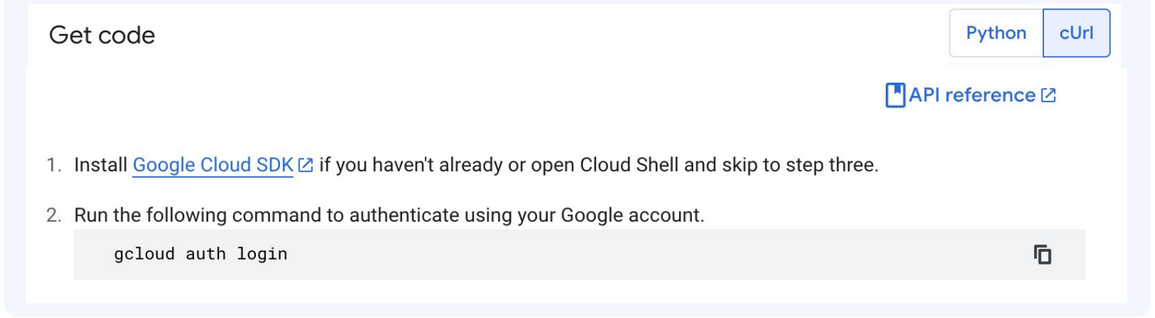
Using SDKs (for example, Java, Python, JavaScript) with a notebook.

APIs

Using APIs with command-line tools.

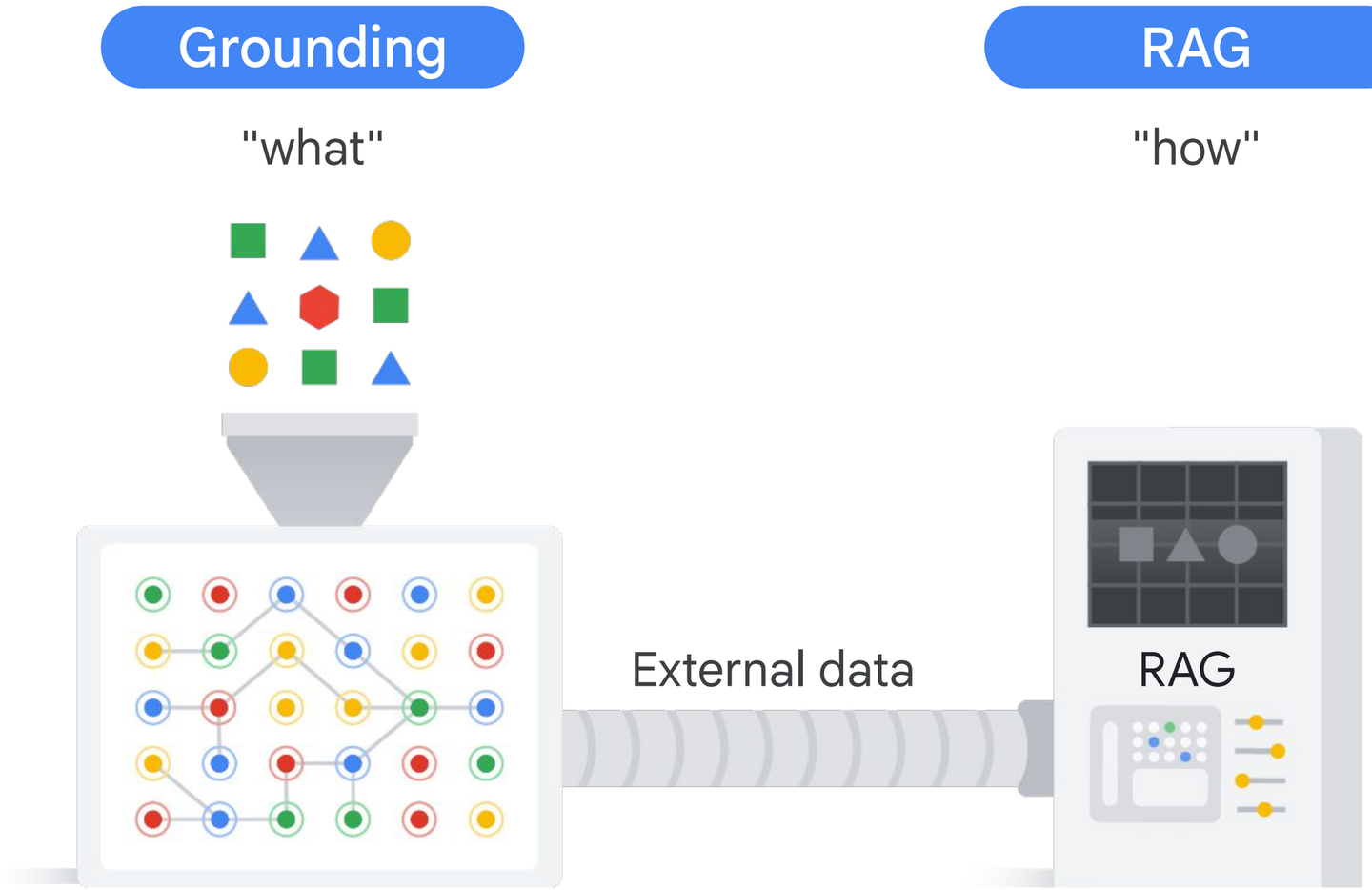






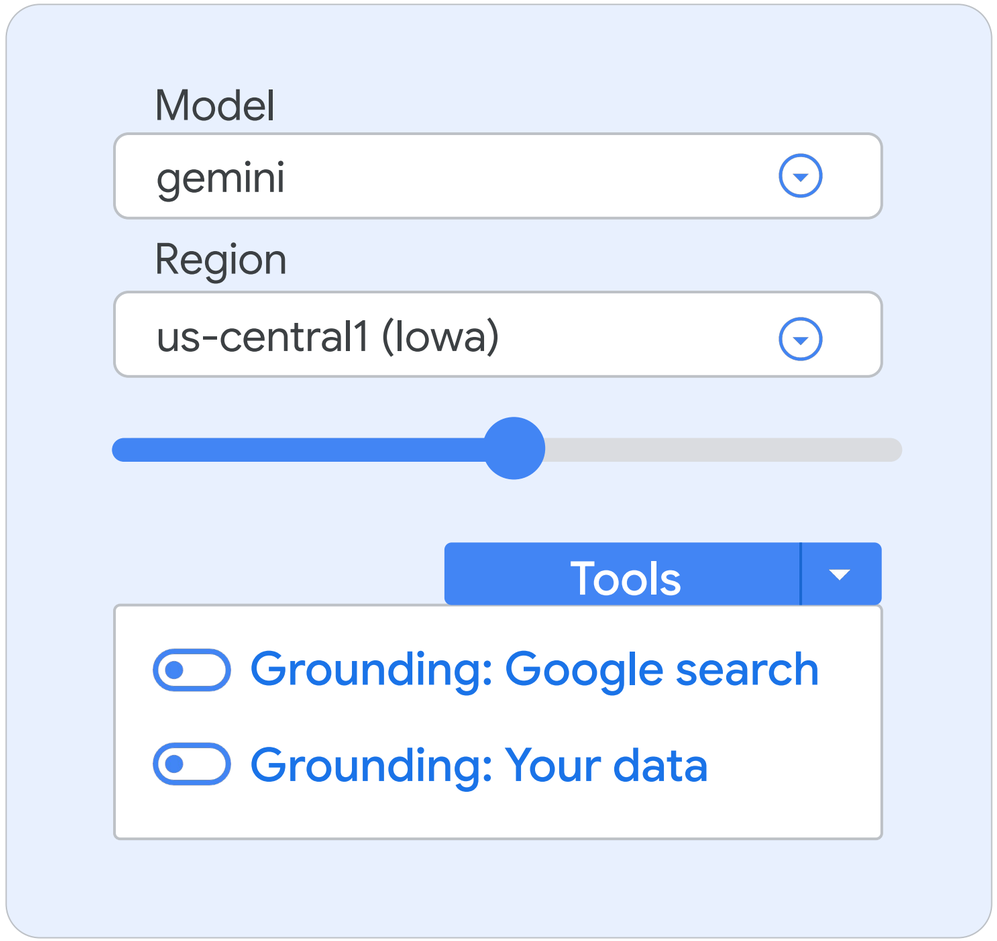
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Grounding and RAG



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Grounding with Agent Studio



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Agent building with Google Cloud

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Applications/Solutions

Gemini Enterprise app

Gemini Enterprise for CX

The AI agent entry point for **business users**

Development

Agent Platform - ADK, Agent Studio, Agents CLI, Agent Garden

The AI agent entry point for **developers**

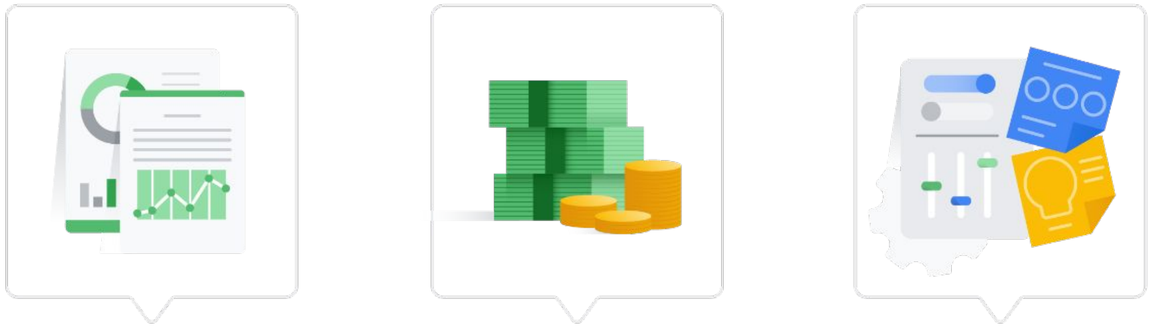
Foundation models

Model Garden - Gemini, Veo, Nano Banana, Embeddings



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Gemini Enterprise app



AI agent AI agent AI agent

- No-code application
- An enterprise platform uniting AI, Google search, and company data
- Multimodal search of the organization data
- A hub of AI agents

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Design → Development → Deployment



Agent Studio

- Low-code visual interface.
- Natural language testing.
- Streamlined design environment.



ADK

- Graph-based logic.
- A2A orchestration.
- Secure workspaces.

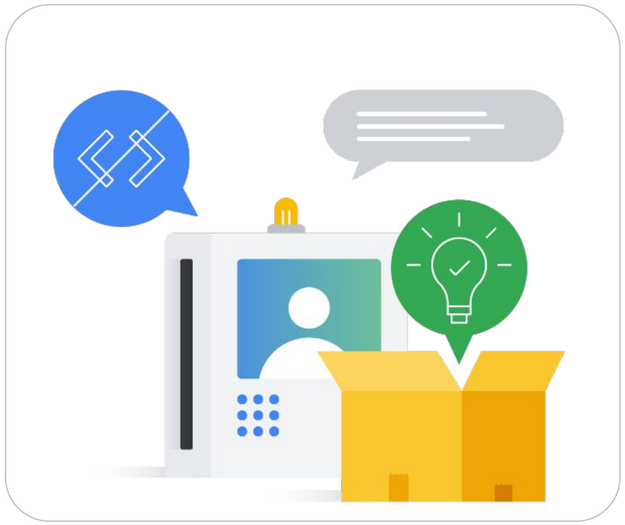


Agent Runtime

- High-performance infrastructure.
- Production-scale management.
- Integrated observability.

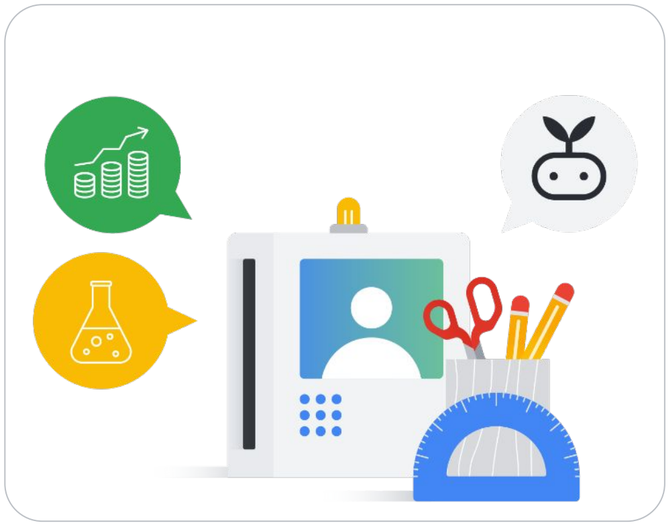
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Navigate the agent development tools



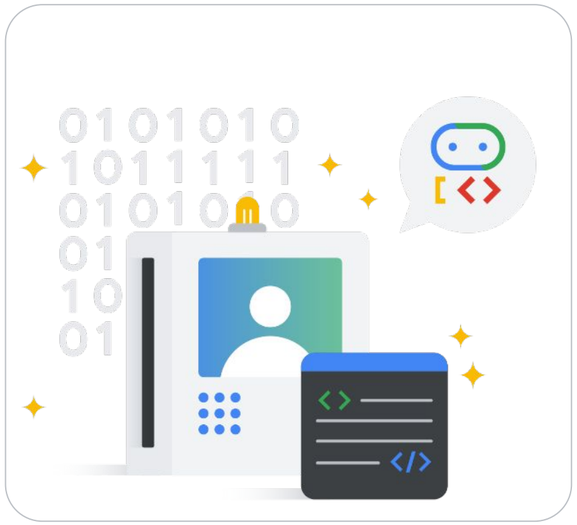
Out-of-box solutions
No code

Gemini Enterprise app
Gemini Enterprise for CX



Customizable template
No/low code

Agent Platform: Agent Studio, Agent Garden



Custom development
Pro code

Agent Platform: ADK